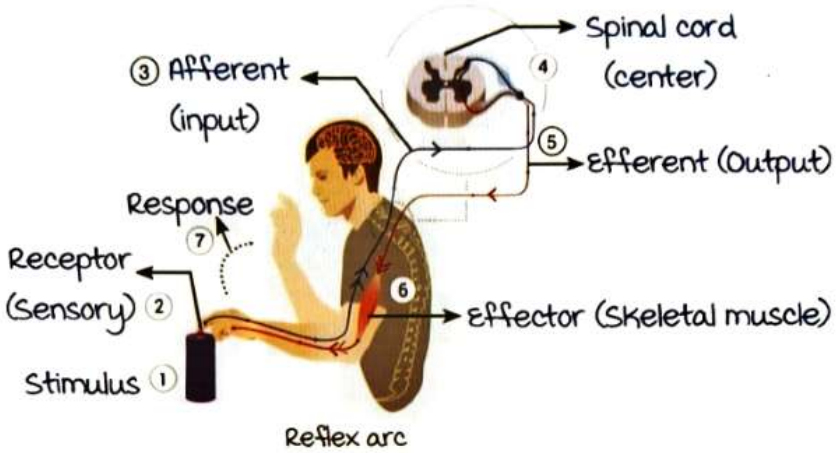
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----- Active space -----

SPINAL REFLEXES



Stretch Reflex

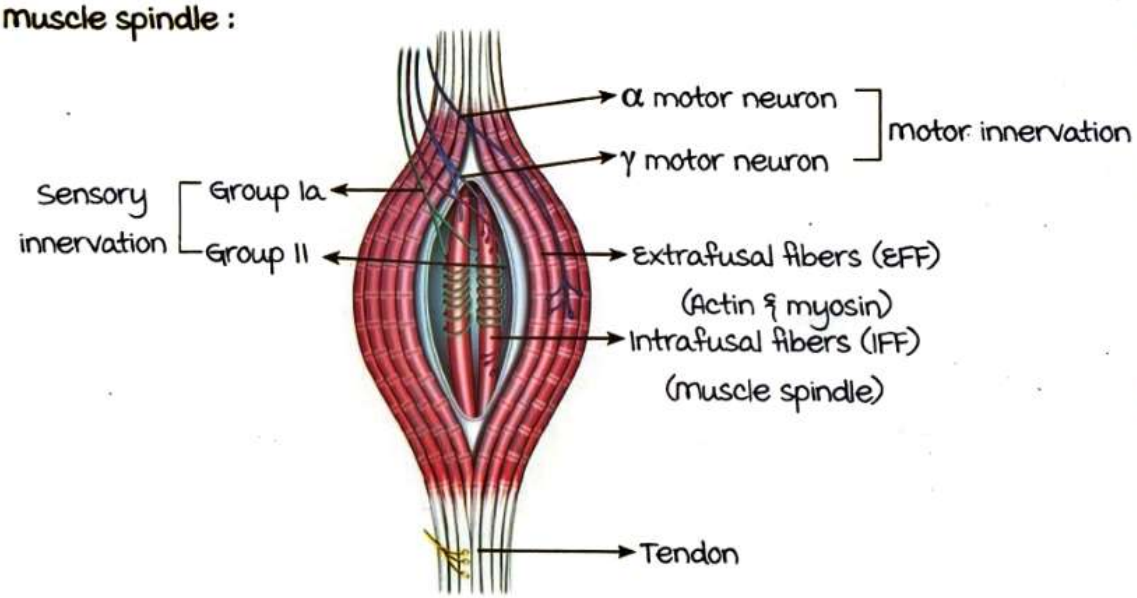
00:04:32

STIMULUS :

Stimulus (Passive stretch) —Induces→ muscle contraction.

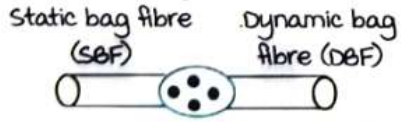
RECEPTOR :

Muscle spindle :



Types of muscle spindle :

1. Nuclear bag fibers (NBF).



2. Nuclear chain fibers (NCF).



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Innervation :

Sensory afferents :

----- Active space -----

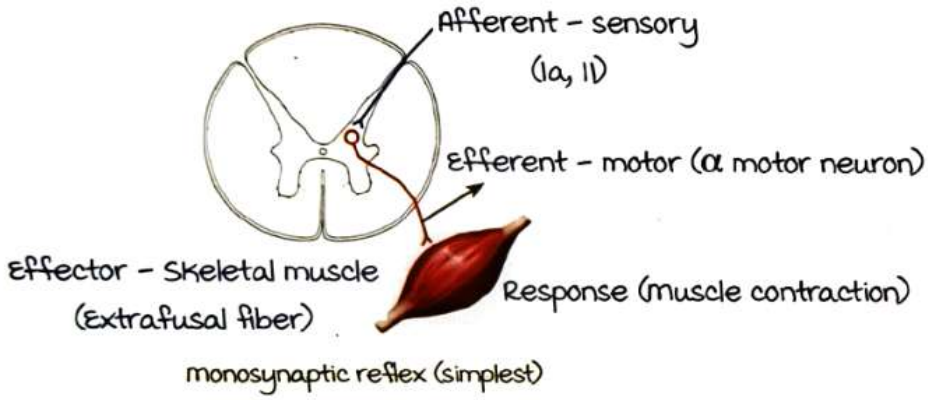
	Group Ia	Group II
AKA	Annulospiral endings Primary	Flower-spray endings Secondary
Innervates	Static bag fiber Dynamic bag fiber NCF	Static bag fiber NCF NCF

motor efferent :

- γ motor neuron.
  Supply intrafusal fibers.
  Types :
  - Dynamic (End plates) : DBF.
  - Static (Trail plates) : SBP, NCF.
- Alpha motor neuron supply extrafusal fibers.

CENTRE :

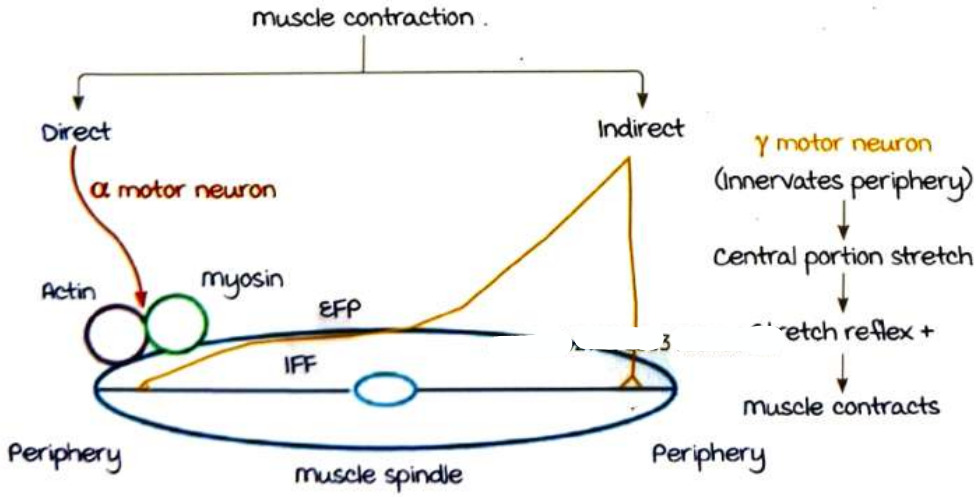
Spinal cord.



monosynaptic reflex (simplest)

RESPONSE :

Alpha Gamma Coactivation :

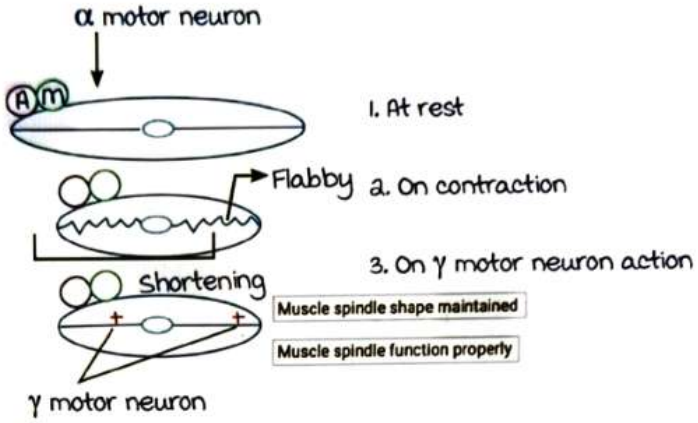


Feedback

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----- Active space -----

Applied aspect :



Jendrassik maneuver :



Clenching teeth

(or)



Interlocking fingers

Followed by

Knee jerk



→ Brisk knee jerk reflex (D/t α & γ motor neuron coactivation)

Inverse Stretch/Relaxation Reflex

00:28:22

- Protective reflex.

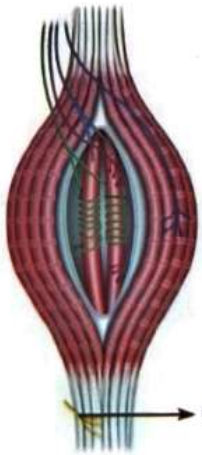
STIMULUS :

Overstretching of muscle spindle.

Eg : Weight lifting.

RECEPTOR :

Golgi tendon organs.



→ Golgi tendon organ afferent

Group Ib (Sensory)

Feedback

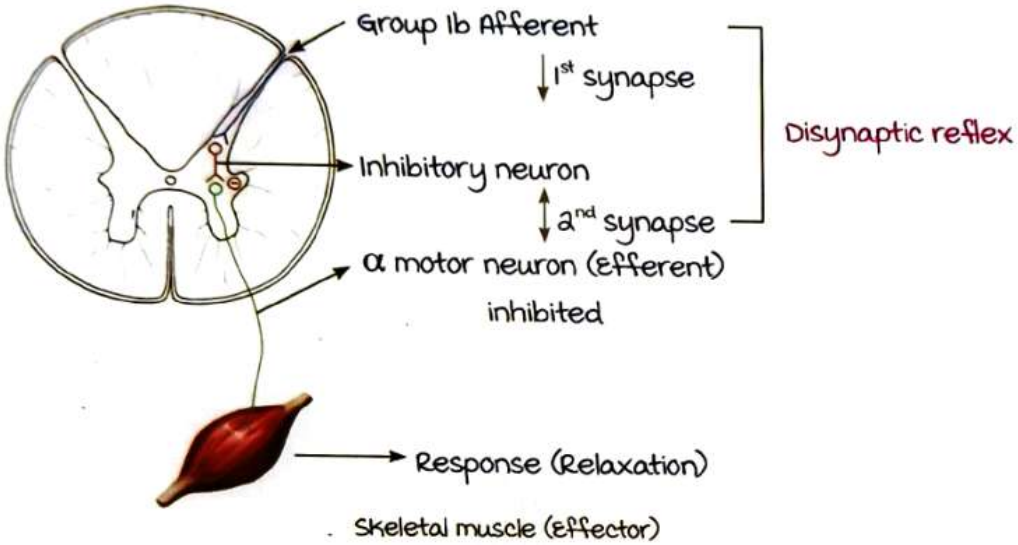
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CENTRE :

Spinal cord.

----- Active space -----



RESPONSE :

Relaxation of muscle.

SUMMARY :

Components	Stretch Reflex	Inverse Stretch Reflex
Stimulus	Stretch	Overstretch
Receptor	muscle spindle (Length sensors)	Golgi tendon organ (Tension sensors)
Sensory input	Ia, II	Ib
Center	Spinal cord	Spinal cord
Motor output	α motor neuron stimulated	α motor neuron inhibited
Response	Contraction	Relaxation
Other names	myotatic reflex	Lengthening reaction
Number of synapses	Monosynaptic	Disynaptic

CLINICAL ASPECTS :

Feedback

Clasp knife spasticity

Initial ↑ in resistance

(D/t stretch reflex)

↓

↓ in resistance (D/t inverse stretch reflex)

Ankle clonus

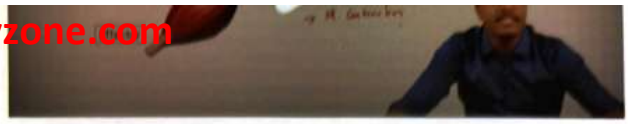
Sudden & sustained stretch

↓

Ankle clonus

(D/t cycle of stretch & inverse stretch reflex)

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----- Active space -----

Other Reflexes

00:42:40

1. WITHDRAWAL REFLEX :

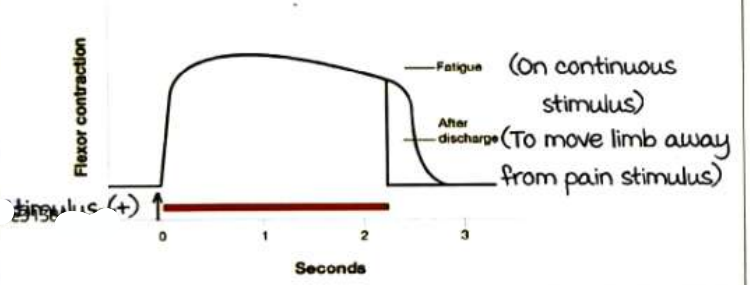
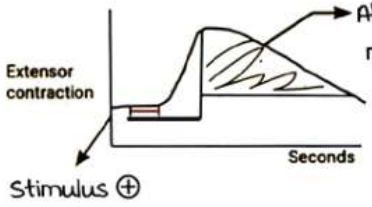


Affected leg : Flexion
(Flexor ⊕, extensor ⊖)

Painful stimulus +

Balancing leg : Extension
(Flexor ⊖, extensor ⊕)

Polysynaptic Reflex

Myogram Flexor Reflex	Myogram Crossed Extensor Reflex
No latency (Immediate, first/earliest response)	Latency (+)
	

(On continuous stimulus)
(To move limb away from pain stimulus)

After discharge (To maintain posture/ balance)

Stimulus ⊕

2. POSITIVE SUPPORTING REACTION / MAGNET REACTION



Foot pad
Pressure
Integration at spinal cord



Limb extension
(Provides support/ balance)

3. PLACING REACTION



Flexion of knee & hip
↓ Integration : Cerebral cortex
For firm placement of foot on table

Dorsum of foot on table

4. RIGHTING REFLEX



Laid on side
↓
Series of body movement
↓ Integration : midbrain
standing

5. HOPPING REFLEX



Standing on two legs
↓
moved horizontally
↓ Integration : Cerebral cortex
Quickly hop (Postural reflex)

6. MASS REFLEX



Paraplegic patient
↓
Mild painful stimuli
↓
Profuse sweating, systolic BP > 200 mg
(D/t activation of multiple spinal segments)
↓
Bowel & bladder evacuation

Feedback